

DUNCAN WOOD

Washington, D.C. Metro Area | duncanwoodpro@gmail.com

Portfolio: duncanwoodpro.netlify.app | LinkedIn: linkedin.com/in/duncanwoodpro | GitHub: github.com/Duncan-Wood

SUMMARY

Product-minded Full-Stack & Web Engineer with nearly 3 years of professional experience owning, scaling, and maintaining high-traffic web applications, custom multi-tenant platforms, and headless CMS architectures. Proven track record as a Lead Developer with collaborative team decision-making, delivering end-to-end web experiences, robust RESTful APIs, and optimized databases using TypeScript, Next.js, React, Ruby on Rails, Python (FastAPI/Flask), and PostgreSQL. Focused on performance optimization, security hardening (credential rotation), and technical SEO/AEO.

TECHNICAL SKILLS

- **Languages & Runtimes:** JavaScript / TypeScript, Ruby on Rails, Python (FastAPI, Flask), Node.js, SQL (PostgreSQL, BigQuery), HTML, CSS.
- **Frontend & CMS Architecture:** Next.js (App Router), React, Contentful, Hotwire (Turbo/Stimulus), Tailwind CSS, UI Component Libraries.
- **APIs & Security:** REST APIs, CSRF/XSS Remediation, Cloudflare Turnstile, Auth/OAuth, API Key Rotation.
- **Cloud & DevOps:** Google Cloud Platform (Cloud Run, Functions, Cloud Tasks, Secret Manager), Vercel, Docker, Git / GitHub Actions (CI/CD), Sentry, PostHog, Scout APM.
- **Testing & Quality Assurance:** RSpec, Capybara, Playwright, pytest.

EXPERIENCE

Software Engineer | EcoMap Technologies, Inc. | November 2023 – July 2026

- **Platform Ownership & Scale:** Served as sole engineer responsible for a 70+ customer, multi-tenant white-labeled platform after its original team departed, hitting formal stabilization goals (escalations under 3/month) while continuing to ship features solo—then owned the engineering side of migrating those customers onto a replacement product and decommissioning the legacy platform.
- **Decommissioning Pipeline:** Built and operated a safety-checked deletion pipeline (export → verify record counts match → delete → poll until confirmed empty, dry-run by default) to decommission sub-applications and databases across 50+ client environments with no data loss.
- **Full-Stack Feature Ownership:** Served as lead developer on an internal data-review tool built from scratch (Next.js, TypeScript, Supabase) to production use, and shipped ~93 merged pull requests over 22 months to the flagship Ruby on Rails platform alongside three full-time platform engineers—including building key analytics report submodules (Population Trends, Business Formation & Survival) by translating BigQuery-sourced federal datasets into interactive user metrics.
- **AI / LLM Integration:** Generalized a colleague's single-report LLM summarization service into a shared concern the whole report suite could use, then integrated three further report modules onto it; migrated a production RAG chatbot's embedding model for an estimated ~5x reduction in embedding costs.
- **Security & Incident Response:** Diagnosed and resolved a production crash resulting from unescaped characters in JSON parsing. Led the credential rotation for four critical services following a third-party security incident, migrating from shared API keys to per-consumer, independently rotatable secrets.
- **Delivery Process & Quality:** Acted as a required code reviewer, authoring test plans and signing off QA cycles across multiple platform releases, while simultaneously delivering feature work.
- **Cost & Performance Optimization:** Resolved a sudden doubling of a third-party search platform (Algolia) bill caused by high-volume bot traffic, implementing multiple rounds of request-filtering and rate-limiting remediations that successfully restored monthly costs to baseline.

Contract Full-Stack Developer | Mighty Crow | January 2025 – November 2025

- **Compliance & Workflow Automation:** Designed a rule-based compliance engine using Xano and WeWeb to automate real-time housing certification cascades across regulatory registries; engineered backend workflows ensuring strict data consistency and state management across legal processes.
- **HIPAA-Compliant Microservice:** Independently designed and deployed a HIPAA-compliant document-conversion microservice (Python/Flask) on GCP Cloud Run. Implemented automated CI/CD deployment, strict encryption controls, and a full automated test suite to replace a commercial alternative.
- **Form Builder & Data Modeling:** Designed and built a general-purpose form-builder feature from a loose spec (supporting form bundling and anonymous public submissions), evaluating and documenting data-modeling tradeoffs and authoring complete technical handover documentation.

PROJECTS

- **my-heartrate (Real-Time Biometric Ingestion & Analysis):** Built a local-first, real-time biometric streaming application that pulls live heart-rate data off a smartwatch over Bluetooth Low Energy (BLE), persists it, and pushes live updates to a browser visualization layer via WebSockets.
- **WordPress-to-Contentful Migration Pipeline:** Engineered an automated headless CMS migration pipeline, writing a Contentful adapter abstraction layer that handles rich-text conversions, inline-image migrations with size correction, and tag deduplication.

EDUCATION

- **General Assembly** — Software Engineering Immersive (Completed April 2023)
- **University of Central Florida** — B.A. in Writing and Rhetoric (Graduated May 2020) · Minor in Diversity and Social Inequality